

Tyler Vander Mooren

CYB 310: Malware Analysis

Professor Troy Adams

February 19, 2026

Lab 3: Basic Dynamic Analysis of Malware

The purpose of this lab was to perform basic dynamic analysis on a malicious document within a controlled virtual machine (VM) environment. Dynamic analysis focuses on observing runtime behavior, which includes process execution, file system modifications, registry activity and network communication. The objective of this lab was to determine if the sample malicious document exhibited behavior consistent with common malware techniques such as exploit-driven code execution, payload delivery, persistence establishment or command-and-control (C2) communication. For this lab, the analyzed sample was `important.pdf`. The name choice uses a vague and urgent style which suggests this could be a potential social engineering lure to prompt user interaction.

Environment Setup

The analysis was done inside a Windows 10 VM running in Oracle VirtualBox. The VM was configured with the following controls:

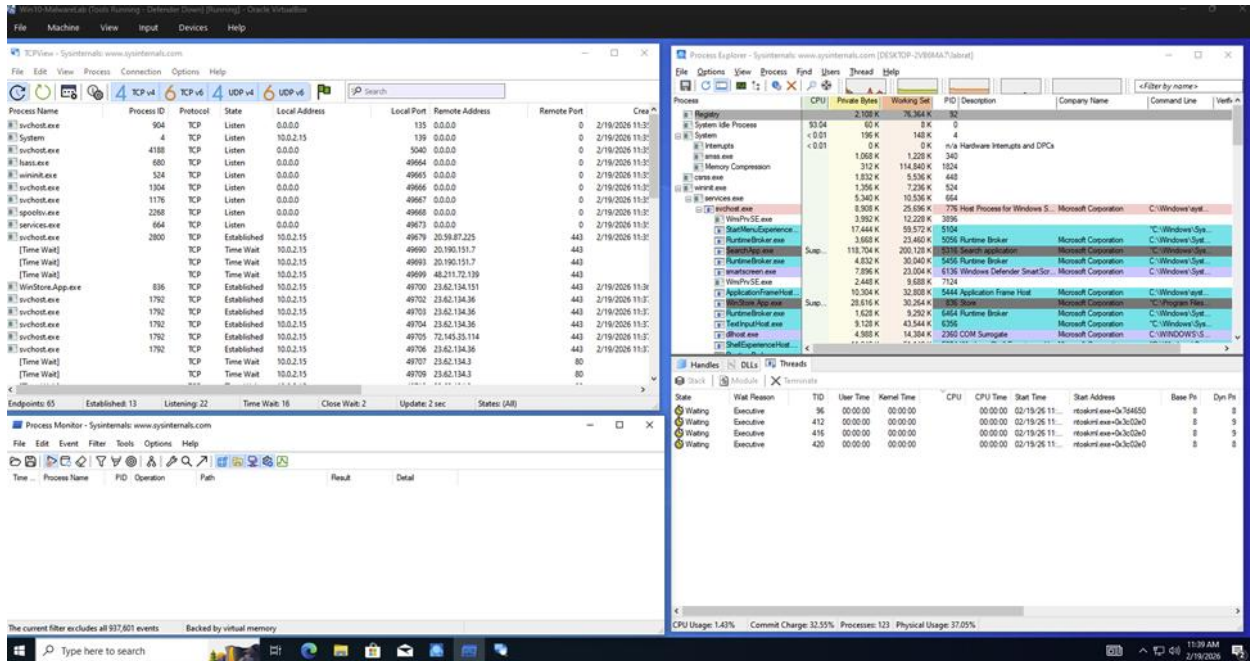
- NAT networking mode
- Shared folders disabled
- Clipboard and drag and drop disabled
- Windows Defender disabled
- Snapshot created prior to execution

Monitoring tools used:

- Process Explorer (process monitoring)
- Process Monitor (File and Registry Monitoring)
- TCPView (Network Monitoring)

As shown in Figure 1, the monitoring environment and the baseline before opening the malicious pdf.

Figure 1. Monitoring Tools Baseline Prior to Execution



Execution and Process Monitoring

During the opening of the file important.pdf, process explorer was used to observe process hierarchy and detect potential child process creation. However, during execution, Adobe Reader launched normally. There was no additional child processes spawned and more specifically there were no instances of any cmd.exe, powershell.exe or wscript.exe behavior observed. CPU usage also remained consistent with normal document rendering behavior. So, the absence of child process creation suggests that no script execution or exploit triggered execution occurred during runtime (see Figure 2).

[illegible]

Process Monitor was able to capture around 900,000 total system events during the observation window. When filtered down for AcroRd32.exe the result was around 100,000 events associated with Adobe Reader. But most of the operations were ReadFile, CreateFile, RegOpenKey, RegQueryValue which are typical Adobe Reader operations. Furthermore, there were no executable files written to user directories such as AppData, Temp or Startup Folder. So, there was no evidence of secondary payload creation and no file-based persistence mechanisms were observed. All file activity stayed consistent with normal PDF behavior.

As mentioned previously, registry monitoring revealed routine registry queries that are associated with Adobe Reader. No instances of RegSetValue or RegCreateKey were observed and additionally, no modifications were detected within common persistence locations such as in

HKCU\...\CurrentVersion\Run, HKLM\...\CurrentVersion\Run. This indicates that the sample did not attempt to establish registry-based persistence.

Network Activity

For network monitoring TCPView and Process Monitor were used to detect any activity. There were no TCP Connect events when the pdf was opened. Also, no DNS resolution attempts were made or outbound HTTP/HTTPS connections were observed from AcroRD32.exe. The complete absence of outbound communication suggests the sample did not attempt to retrieve a secondary payload or establish C2 communication.

Correlation with Expected Malware Behavior

Some of the expected behaviors from a PDF-based malware are exploiting vulnerabilities in Adobe Reader to achieve code execution, spawning system utilities, deploying secondary payloads to disk, establishing persistence through registry modification and initiating outbound C2 communications. None of these behaviors were observed during dynamic analysis of important.pdf. The absence of exploit-triggered behavior suggests now that either the sample targets a vulnerable version of Adobe Reader, needs additional user interaction to activate malicious content or it is a non-functional or benign document. The observed behavior does not align with any typical exploit-based PDF malware patterns.

Dynamic analysis of the PDF sample important.pdf revealed no observable malicious behavior within the controlled Windows 10 VM environment. Monitoring tools confirmed the absence of abnormal process creation, file system modification, registry persistence mechanisms and outbound network communications. While the filename does suggest potential social engineering intent, runtime behavior did not support the presence of active exploit or payload functionality. However, these findings do highlight the important limitation of dynamic analysis.

Malware behavior can be heavily dependent on environmental conditions. As a result, the absence of observable malicious behavior does not definitively confirm that the document is benign, but rather that no malicious functionality was activated under the controlled environment.